Limits

Questions 1–11

Question 1. Find

a) $\lim_{x\to\infty}(1+e^{-x})$

b) $\lim_{x\to 0}(x^4e^x)$

Question 2. Evaluate each of the following:

a) $\lim_{x\to 0}\sqrt{\dfrac{x^3-2x^2+10}{4x^3+1}}$

b) $\lim_{x\to 0}\dfrac{x+4}{\sqrt{3x^2-5x+2}}$

c) $\lim_{x\to 0}\dfrac{\sqrt{2x^3-5x^2-1}}{x^2-2}$

Question 3. Find each of the following:

a) $\lim_{n\to\infty} \dfrac{5-4n}{n}$

b) $\lim_{n\to\infty} \dfrac{2n-3}{n^2}$

c) $\lim_{n\to\infty} \dfrac{n+7}{2n+1}$

d) $\lim_{n\to\infty} \dfrac{10-3n}{n-3}$

Question 4. Evaluate these limits:

a) $\lim_{n\to\infty} \dfrac{2n^2-4n+5}{3n^2+n-4}$

b) $\lim_{n\to\infty} \dfrac{5n^2+2n+1}{2n^2-n-7}$

c) $\lim_{n\to\infty} \dfrac{5n^2-4n+5}{4-n-7n^2}$

Question 5. Find each of the following:

a) $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2}$

b) $\lim_{x \to 1} \dfrac{x^3 - 1}{x - 1}$

Question 6. Find each of the following:

a) $\lim_{x \to 3} \dfrac{\sqrt{x} - \sqrt{3}}{x - 3}$

b) $\lim_{x \to 4} \dfrac{\sqrt{x} - 2}{x - 4}$

c) $\lim_{x \to k} \dfrac{\sqrt{k} - \sqrt{x}}{x - k}$, for positive constant k

Question 7. Use small angle approximations to show that

a) $\lim_{x\to 0}\left(\frac{\sin x}{x}\right) = 1$

b) $\lim_{x\to 0}\left(\frac{\cos x - 1}{x}\right) = 0$

Question 8. Find each of the following:

a) $\lim_{x\to 0}\left(\frac{\sin 3x}{x}\right)$

b) $\lim_{x\to 0}\left(\frac{1-\cos 4x}{x^2}\right)$

c) $\lim_{x\to 0}\left(\frac{4\cos x}{\sin x + \cos x}\right)$

Question 9. By using the approximation $\ln(1+\frac{x}{2}) = \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{24}$ where x is small, find $\lim_{x\to 0}\left(\frac{8\ln(1+\frac{x}{2})}{x}\right)$.

Question 10. By using the approximations $\cos 2x \approx 1 - 2x^2$ and $\sin 3x \approx 3x - \frac{9}{2}x^3$ for small values of x, find $\lim_{x\to 0}\left(\frac{3x\cos 2x - \sin 3x}{3x^3}\right)$.

Question 11. By considering the first 4 terms in the expansion of $\sqrt{4+x}$, find $\lim_{x\to 0}\left(\dfrac{-2+\sqrt{x+4}}{x}\right)$.